

$$q_e \frac{1}{2} i2_{-1} i3_{-1} i4_{-1} \rightarrow$$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{216} \frac{1}{m_e^2} (s_Y^2 \overline{y_e}^{pi3} y_e^{pi4} (-54 c_Y^2 \overline{y_u}^{i2r} y_u^{i1r} + 5 g_1^2 \delta_{i1i2}) + 13 g_1^2 c_Y^2 \overline{y_u}^{i2p} y_u^{i1p} \delta_{i3i4} - \right. \\ & \quad s_Y^2 \overline{y_d}^{i2p} y_d^{i1p} (54 c_Y^2 \overline{y_e}^{ri3} y_e^{ri4} + 5 g_1^2 \delta_{i3i4})) - \frac{2}{81} \sum_p g_1^4 LF_{3,0}[m_d^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{108} \sum_p g_1^4 LF_{4,-1}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \frac{8}{405} \sum_p g_1^4 LF_{5,-2}[m_d^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{2}{27} \sum_p g_1^4 LF_{3,0}[m_e^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{36} \sum_p g_1^4 LF_{4,-1}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{8}{135} \sum_p g_1^4 LF_{5,-2}[m_e^p] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} \sum_p g_1^4 LF_{3,0}[m_l^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{72} \sum_p g_1^4 LF_{4,-1}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \frac{4}{135} \sum_p g_1^4 LF_{5,-2}[m_l^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{81} \sum_p g_1^4 LF_{3,0}[m_q^p] \delta_{i1i2} \delta_{i3i4} + \frac{5}{216} \sum_p g_1^4 LF_{4,-1}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{405} \sum_p g_1^4 LF_{5,-2}[m_q^p] \delta_{i1i2} \delta_{i3i4} - \frac{8}{81} \sum_p g_1^4 LF_{3,0}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{27} \sum_p g_1^4 LF_{4,-1}[m_u^p] \delta_{i1i2} \delta_{i3i4} - \frac{32}{405} \sum_p g_1^4 LF_{5,-2}[m_u^p] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{18} (s_Y^2 \overline{y_e}^{pi3} y_e^{pi4} (-9 c_Y^2 \overline{y_u}^{i2r} y_u^{i1r} + g_1^2 \delta_{i1i2}) + 4 g_1^2 c_Y^2 \overline{y_u}^{i2p} y_u^{i1p} \delta_{i3i4} - \\ & \quad s_Y^2 \overline{y_d}^{i2p} y_d^{i1p} (9 c_Y^2 \overline{y_e}^{ri3} y_e^{ri4} + 2 g_1^2 \delta_{i3i4})) LF_{1,2}[m_\Phi] + \\ & \quad \frac{1}{36} (s_Y^2 \overline{y_e}^{pi3} y_e^{pi4} (9 c_Y^2 \overline{y_u}^{i2r} y_u^{i1r} + g_1^2 \delta_{i1i2}) - 3 g_1^2 c_Y^2 \overline{y_u}^{i2p} y_u^{i1p} \delta_{i3i4} + \\ & \quad 3 s_Y^2 \overline{y_d}^{i2p} y_d^{i1p} (-3 s_Y^2 \overline{y_e}^{ri3} y_e^{ri4} + g_1^2 \delta_{i3i4})) LF_{2,1}[m_\Phi] - \\ & \quad \frac{1}{108} g_1^2 (3 s_Y^2 \overline{y_e}^{pi3} y_e^{pi4} \delta_{i1i2} + (9 s_Y^2 \overline{y_d}^{i2p} y_d^{i1p} - 9 c_Y^2 \overline{y_u}^{i2p} y_u^{i1p} + 4 g_1^2 \delta_{i1i2}) \delta_{i3i4}) \\ & \quad LF_{3,0}[m_\Phi] + \frac{5}{72} g_1^4 LF_{4,-1}[m_\Phi] \delta_{i1i2} \delta_{i3i4} - \frac{4}{135} g_1^4 LF_{5,-2}[m_\Phi] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{27} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{18} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{135} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{36} g_1^4 LF_{2,1,0}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 LF_{2,2,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{18} g_1^4 LF_{3,1,-1}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 LF_{4,1,-2}[m_1, m_e^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{36} g_1^4 LF_{2,1,0}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 LF_{2,2,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{18} g_1^4 LF_{3,1,-1}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 LF_{4,1,-2}[m_1, m_e^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{1296} g_1^4 LF_{2,1,0}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{2,2,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{648} g_1^4 LF_{3,1,-1}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{4,1,-2}[m_1, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{1296} g_1^4 LF_{2,1,0}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{2,2,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{648} g_1^4 LF_{3,1,-1}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{4,1,-2}[m_1, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{48} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{24} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{48} g_1^2 g_2^2 LF_{2,1,0}[m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{2,2,-1}[m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{24} g_1^2 g_2^2 LF_{3,1,-1}[m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{4,1,-2}[m_2, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{2,1,0}[m_d^p, \tilde{\mu}] \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{2,2,-1}[m_d^p, \tilde{\mu}] \delta_{i3i4} + \\ & \quad \frac{1}{18} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{3,1,-1}[m_d^p, \tilde{\mu}] \delta_{i3i4} + \frac{1}{18} g_1^4 LF_{2,1,0}[m_e^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{36} g_1^4 LF_{3,1,-1}[m_e^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^4 LF_{2,1,0}[m_e^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{36} g_1^4 LF_{3,1,-1}[m_e^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y_e}^{pi3} y_e^{pi4} LF_{2,1,0}[m_l^p, \tilde{\mu}] \delta_{i1i2} - \\ & \quad \frac{1}{36} g_1^2 \overline{y_e}^{pi3} y_e^{pi4} LF_{2,2,-1}[m_l^p, \tilde{\mu}] \delta_{i1i2} - \frac{1}{36} g_1^2 \overline{y_e}^{pi3} y_e^{pi4} LF_{3,1,-1}[m_l^p, \tilde{\mu}] \delta_{i1i2} + \\ & \quad \frac{1}{648} g_1^4 LF_{2,1,0}[m_q^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{3,1,-1}[m_q^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{24} g_1^2 g_2^2 LF_{2,1,0}[m_q^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{3,1,-1}[m_q^{i1}, m_2] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_q^{i1}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_q^{i1}, m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{648} g_1^4 LF_{2,1,0}[m_q^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \frac{1}{1296} g_1^4 LF_{3,1,-1}[m_q^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{1}{24} g_1^2 g_2^2 LF_{2,1,0}[m_q^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} - \frac{1}{48} g_1^2 g_2^2 LF_{3,1,-1}[m_q^{i2}, m_2] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_q^{i2}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_q^{i2}, m_3] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{2}{9} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} LF_{2,1,0}[m_u^p, \tilde{\mu}] \delta_{i3i4} - \frac{1}{9} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} LF_{2,2,-1}[m_u^p, \tilde{\mu}] \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} LF_{3,1,-1}[m_u^p, \tilde{\mu}] \delta_{i3i4} + \frac{1}{18} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{2,1,0}[\tilde{\mu}, m_d^p] \delta_{i3i4} + \\ & \quad \frac{5}{36} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{3,1,-1}[\tilde{\mu}, m_d^p] \delta_{i3i4} - \frac{1}{36} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{4,1,-2}[\tilde{\mu}, m_d^p] \delta_{i3i4} - \\ & \quad \frac{1}{36} g_1^2 \overline{y_e}^{pi3} y_e^{pi4} LF_{2,1,0}[\tilde{\mu}, m_l^p] \delta_{i1i2} + \frac{5}{36} g_1^2 \overline{y_e}^{pi3} y_e^{pi4} LF_{3,1,-1}[\tilde{\mu}, m_l^p] \delta_{i1i2} - \\ & \quad \frac{1}{18} g_1^2 \overline{y_e}^{pi3} y_e^{pi4} LF_{4,1,-2}[\tilde{\mu}, m_l^p] \delta_{i1i2} - \frac{1}{9} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} LF_{2,1,0}[\tilde{\mu}, m_u^p] \delta_{i3i4} - \\ & \quad \frac{1}{36} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} LF_{3,1,-1}[\tilde{\mu}, m_u^p] \delta_{i3i4} - \frac{1}{36} g_1^2 \overline{y_u}^{i2p} y_u^{i1p} LF_{4,1,-2}[\tilde{\mu}, m_u^p] \delta_{i3i4} - \\ & \quad \frac{1}{72} g_1^4 LF_{2,1,1,-1}[m_1, m_e^{i3}, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_e^{i3}, m_q^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{72} g_1^4 LF_{2,1,1,-1}[m_1, m_e^{i4}, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{36} g_1^4 m_1^2 LF_{2,1,1,0}[m_1, m_e^{i4}, m_q^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{4} \overline{y_d}^{i2p} y_d^{i1p} \overline{y_e}^{ri3} y_e^{ri4} LF_{2,1,1,-1}[\tilde{\mu}, m_d^p, m_l^r] - \\ & \quad \frac{1}{2} \tilde{\mu}^2 \overline{y_e}^{pi3} y_e^{pi4} \overline{y_u}^{i2r} y_u^{i1r} LF_{2,1,1,0}[\tilde{\mu}, m_l^p, m_u^r] \Big) \end{aligned}$$